

1. Set-up

Input X (4×4), kernel K (2×2), stride = 1, no padding.

X			
1	2	3	0
0	1	2	1
1	0	1	2
2	1	0	1

K	
1	0
-1	1

Output size = $(4 - 2)/1 + 1 = 3 \Rightarrow Y$ is 3×3

$$Y(i, j) = \sum_m \sum_n X(i+m, j+n) \cdot K(m, n) = X(i, j) \cdot 1 + X(i, j+1) \cdot 0 + X(i+1, j) \cdot (-1) + X(i+1, j+1) \cdot 1$$

2. Nine kernel positions (patch $\rightarrow$ products $\rightarrow$ sum)

$Y(0,0)$: patch $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \rightarrow 1(1) + 2(0) + 0(-1) + 1(1) = 1 + 0 + 0 + 1 = 2$

$Y(0,1)$: patch $\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \rightarrow 2(1) + 3(0) + 1(-1) + 2(1) = 2 + 0 + (-1) + 2 = 3$

$Y(0,2)$: patch $\begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} \rightarrow 3(1) + 0(0) + 2(-1) + 1(1) = 3 + 0 + (-2) + 1 = 2$

$Y(1,0)$: patch $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \rightarrow 0(1) + 1(0) + 1(-1) + 0(1) = 0 + 0 + (-1) + 0 = -1$

$Y(1,1)$: patch $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \rightarrow 1(1) + 2(0) + 0(-1) + 1(1) = 1 + 0 + 0 + 1 = 2$

$Y(1,2)$: patch $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \rightarrow 2(1) + 1(0) + 1(-1) + 2(1) = 2 + 0 + (-1) + 2 = 3$

$Y(2,0)$: patch $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \rightarrow 1(1) + 0(0) + 2(-1) + 1(1) = 1 + 0 + (-2) + 1 = 0$

$Y(2,1)$: patch $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \rightarrow 0(1) + 1(0) + 1(-1) + 0(1) = 0 + 0 + (-1) + 0 = -1$

$Y(2,2)$: patch $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \rightarrow 1(1) + 2(0) + 0(-1) + 1(1) = 1 + 0 + 0 + 1 = 2$

3. Output feature map

Y (3×3)		
2	3	2
-1	2	3
0	-1	2

4. Check

`scipy.signal.correlate2d(X, K, mode='valid')` gives the same 3×3 matrix. (Note: CNNs use cross-correlation — the kernel is not flipped. True convolution with the flipped kernel $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ would give $\begin{bmatrix} 0 & 1 & 4 \\ -1 & 0 & 3 \\ 2 & -1 & 0 \end{bmatrix}$.)